

Prashant Patel

A solution oriented **System Architect/Product Owner** and **Certified Scrum Master™** practitioner with 18+ years of experience in design & development of versatile product domains like Cloud application, NMS, Fiber Technology, Carrier Ethernet, Mobile Technology and Embedded systems with excellent programming knowledge in Python and C/C++ along with exposure of managing large team members.

EXPERIENCE SUMMARY

Dixon Electro Appliances Pvt Ltd, Gurugram — Assistant General Manager (Telecom)

OCT 2022 - PRESENT

Worked on integration of TR-069(CWMP) and secret password enhancement in openWRT for Xstream Fiber product.

Capgemini(Adtran Networks), Hyderabad — System Architect/Product Owner/Program Manager

FEBRUARY 2013 - OCT 2022

Worked on multiple products of layer 2 switches based on Gfast Technology, Fiber Technology OLT(GPON, XGS PON) & ONT(602X, IBONT), Carrier Ethernet TA5K(LAG, Cross slot LAG, ERPS) to Network Management System product like PMA (Persistence Monitoring Agent) and MCP(Mosaic Cloud Platform) plugins which control hybrid protocol (Netconf, Restconf, SNMP, CLI etc) devices.

CMC India Pvt Ltd, Hyderabad — Senior IT Engineer

MARCH 2010 - JANUARY 2013

Worked on development of new CLI commands for multiple access module cards like RPR and OGE cards. Also worked on static LAG(Link Aggregation Group) for OGE cards where multiple ports can be bundled together to pass the higher data without changing hardware capabilities.

EXFO India Pvt Ltd, Pune — Senior Software Engineer

JANUARY 2009 - FEBRUARY 2010

Worked on application of H.323 protocol stack and developed the server application which can be used in the field to test the bulk clients connection with full audio and video payloads.

L&T Infotech Pvt Ltd, Bangalore — Senior Software Engineer

APRIL 2006 - DECEMBER 2008

Worked on Qualcomm mobile handset and QTV server which uses the Real Time Streaming Protocol(RTSP) to establish the session between client and server for carrying the audio/video packets through RTP/RTCP protocols and later play the audio and video packets on device based on timestamp.

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TECHNICAL SKILLS

Programming Languages: Python, C++, C, Data Structure, Multithreading, IPCs, Socket, Java Script etc.

Cloud Platform: Microservices (Plugin), Docker, Kubernetes etc.

DevOps: Pytest, Robot, GTest, Jenkins, CI/CD, Groovy etc.

Debugging Tools: MULTI, JTAG, GDB etc.

Defect & Version Control: Jira, Github, P4v, CSV, Confluence etc.

Protocol and Standards: NetConf, RestConf, YANG Model(JSON/ XML), SIP, H.323, RTSP, SNMP, LAG, ERPS etc.

CERTIFICATIONS

SAFe 4 Certified Scrum Master from Scaled Agile Inc.

SAFe 4 Certified Practitioner from Scaled Agile Inc.

EDUCATION

United College of Engineering and Research, Allahabad — B.Tech(I.T.)

JULY 1999 - OCT 2003

Jawahar Navodaya Vidyalaya, Mirzapur — Science

JULY 1996 - MAY 1997

Aelix Communication System, Bangalore — Member of Technical Staff

MARCH 2005 - MARCH 2006

Worked on a VoIP Phone called 'eNterprise' where I was involved in development of Subscriber and Notification of SIP messages which notify the caller based on their event registration like Date and Time.

PROJECT DETAILS

Project Name	<i>Xstream Fiber ONT</i>
Company	Dixon Electro Appliances Pvt Ltd., Gurugram
Duration	Oct 2022 To Till date
Position	Assistant General Manager
Technology	openWRT, Docker
Summary	New Xstream Fiber ONT supports the new openWRT(open Wireless Router) based code along with GPON stack. The openWRT is based on a package system which provides more freedom to choose vendors and enhance the current functionalities.
Responsibilities	• Integrated the TR-069(CPE WAN Management Protocol) package into openWRT for remote accessing of ONT devices through ACS(Automatic Configuration Server). • Remove the hardcoded password from code with docker secrets functionality. • Higher and ramp up the team members.

Project Name	<i>Gfast Plugin(Mosaic Cloud Platform)</i>
Company	Capgemini(Adtran), Hyderabad
Duration	Dec 2021 To Oct 2022
Position	System Architect/Product Owner/Program Manager
Technology	Python/Pytest and Robot Framework(CI/CD), Netconf, Restconf, Yang
Summary	Gfast Plugin is a Python based device control microservices of MCP(Mosaic Cloud Platform) which handles the request and response from Gfast switches on a very large scale. G.FAST switches are based on CALL HOME functionality of Netconf protocol. Gfast Plugin is used to configure and retrieve the status and performance monitoring data and also supports FCAPS functionalities.
Responsibilities	• Play a System Arch role to lay down the architectural runway for the team for next upcoming design changes in MCP(Gfast Plugin). • Design and developed Fast Reflow concept based on RestConf(edit-config) to reduce the larger chunk of time during push back the original configurations in the software upgrade process. • Play a Product Owner role as content authority to define team backlog prioritizes, user stories, product roadmap etc. • Involved in the planning, execution, refinement and review of each sprint. • Involved in defining cohesive vision with the team by inspecting and evaluating product progress through each iteration. • Involved in code review and demoing the features at system level. • Involved in hiring of new team members and bringing them on board in a short span of time with appropriate ramp-up planning.

Project Name	<i>Fiber Agile Release Train(OLT and TA5K Plugins)</i>
Company	Adtran Networks India Pvt. Ltd, Hyderabad
Duration	July 2017 To July 2021
Position	Product Owner/Senior Staff Engineer
Technology	C++, Python/Pytest and Robot Framework(CI/CD), OLT
Summary	MCP(Mosaic Cloud Platform) has three layers, Presentation, Orchestration and Device Control(Plugins Microservices). By using the Plugins, Configuration and Statistics of devices can be pushed or retrieved through hybrid protocol(RESTConf, NetConf, SNMP, CLI) which is managed by MCP.
Responsibilities	• Played the critical role of Product Owner and delivered end to end solutions for SDX OLT products(GPON, XGS PON). • Involved in development of multiple features of SDX OLT Plugin solution like Ethernet OAM, Yang Data Store Validation, Robot framework for end to end testing, Inspect pages etc. • Involved into the development and delivery of E-Line configuration features through MCP Plugin, which can be established through TA5K, OLT and ONT cards. • Involved in hiring of new team members.

Project Name	<i>PMA(Persistence Monitoring Agent)</i>
Company	Adtran Networks India Pvt Ltd, Hyderabad
Duration/ Position	January 2016 To July 2017/Lead engineer
Technology	C++, Python, Java, Netconf, JSON, XML and CentOS
Summary	The PMA (Persistence Monitoring Agent) is CLI and Web based agent which is used to provision startup and running configuration into DPU (Distributed Point Units).DPU is g.fast based technology switches which need an external agent (PMA) for configuration and performance statistics. PMA designs have majorly two parts one is northbound (upper interface) which is based on RESTCONF protocol and southbound (lower interface) supports the NETCONF protocol which interacts with DPU based on CALL HOME feature.
Handled Module	PAM Proxy, PMA Admin etc.
Responsibilities	• Involved in design, development and Integration testing of many new features on PMA like Diagnostic Job, Backup and Restore Job, Scheduling of Maintenance Job, PM Data Mirroring Job, PM Upload Job etc. • Integrated the valgrind (callgrind) and JIP (Java Interactive Profile) tool into PMA for the evaluation of code performance.

Project Name	<i>TA5K(Total Access 5000 Switch Modules)</i>
Company	Adtran Networks India Pvt Ltd, Hyderabad
Duration/ Position	February 2013 To January 2016/Lead engineer
Technology	C++, JavaScript, MULTI debugger, Adtran OS

Summary	Switch Module card is core of layer 2 switching concept, which is used to process the different types of packets (Ethernet, fiber, DSL etc.) based on VLAN id and MAC addresses. Switch Module is sited in TA5000 Chassis and supports the redundant card feature. If any failure happens in the Active SM card, then the Standby card will take the responsibility within a few seconds with minimal data traffic loss. Switch module card can support up to 22 different Access Modules and handle the all switching of ingress and egress data traffic packets which were received either through backplane or faceplate ports. TA5K chassis also have SCM (Switch Control Module) card, which is used to manage the configuration on different cards.
Handled Module	CLI, SNMP, Web GUI interface and Mocha framework (used to verify GUI pages).
Responsibilities	- Involved in design and development of upgrade and downgrade of gig and ten-gig port interfaces for OSP (OutSide Plant) switch. - Involved in enhancement of ERPS feature in switch module. Like support of the S-tag feature and Topology discovery. - Involved in the design and development of LAG cross slot features across different switches. - Involved in design and development of redundancy support between two different types of SM cards which support either Gig or Ten Gig interfaces. - Code porting for new SM card (1187031F1). - Added Shaper support for Ten Gig interface in SM and implemented the Web GUI page (JavaScript) for customer interface. - Involved in fixing defects/bugs in SM cards.

Project Name	<i>TA5K CLI (Octal Giga Ethernet and Resilient Protection Ring Card)</i>
Company/Client	CMC, Hyderabad/Adtran, Mountain View(CA)
Duration/Position	March 2010 To January 2013/Senior IT Engineer
Technology	C++, MULTI debugger, Adtran OS
Summary	This project was based on the development of new CLI commands for RPR (Resilient Packet Ring) cards based on existing menus. Since, existing menus were operationally quite heavy and slow. That's the reason it was decided to develop a new CLI command for RPR and other similar products in Adtran. The OGE card is an Access Module card which sits in TA5000 chassis, and supports the transmission of 8 GB Ethernet traffic through 8 ports. It supports the host standby redundancy feature through Remote DB feature.
Handled Module	LAG, EVC, EVC-Map and Subtended-Host components of OGE cards.
Responsibilities	- Involved in development of different features on OGE cards like LAG (Link Aggregation), LACP (Link Aggregation Control Protocol) and Redundancy. - Involved in design and development of OGE/RPR CLI commands from existing menus. - Involved in fixing defects/bugs in OGE/RPR cards.

Project Name	<i>H323 Flex Server</i>
Company	EXFO, Pune
Duration/Position	January 2009 to Feb 2010/Senior Software Engineer
Technology	C++, Linux and H.323 (ITU)

Summary	H323 Flex project was based on H.323 protocol (ITU) which provides the flexibility to test the H.323 protocol base devices and it was able to handle the multiple calls and registrations with gatekeeper. The H323 Flex supports the slow call, fast start, tunneling and parallel mode with different types of media codec handling. The objective of the project was to generate 32000 end users with a speed of 200/300 cps with proper statistics of success, failure, skipped and a lot more information about calls, registration and media codec. By these statistics users can find the performance of their devices with different combinations of call modes, with or without registration and media test suites for different codec. This project is part of Navtel IW (Inter Watch) which provides different VoIP protocol support like SIP, MEGACO, HSS and BGF etc.
Handled Module	H225 call signaling and H245 call control (Media negotiation) modules.
Responsibilities	- Involved in the design and development of User Agent side of H.323 protocol for generating and handling multiple calls, registration with gatekeeper. - Involved in design and coding for media negotiation of different types of codec, User Input support for DTMF and RFC2833 for telephone-event handling. - Involved in the fixing of defects/bugs for different customers.

Project Name	<i>QTV Packet Switch Streaming Client/Server</i>
Company/Client	L & T InfoTech, Bangalore/ Qualcomm, Hyderabad
Duration/ Position	April 2006 To Dec 2008/Senior Software Engineer
Technology	C++ and Real Time Streaming Protocol (RFC 2326)
Summary	Packet Switched Streaming (PSS) is a way to stream audio/video media content to mobile devices without storing content on the client. The underlying technology uses RTSP/SDP protocol for signaling and RTP/RTCP for data transfer/control, based on 3GPP recommendation for PSS Client. QTV Server was an internal test server used to test the ability of different streaming scenarios from the Qtv client(mobile handset). Before this there was no flexibility to change the commercial servers to simulate different streaming scenarios for testing. Therefore, if customer issues are reported from the test field then it is hard to debug. With the Qtv Server, it was possible to simulate such scenarios in the lab by replaying captured signaling and data traffic in wireshark (collected in the test field), which makes debugging much easier. There was lots of flexibility in the server to reproduce different complicated or specific scenarios based on customer requirements.
Handled Module	QTV streaming
Responsibilities	- Involved in development of RTSP features which were needed by different clients and also supported the enhancement of other features in RTSP. - The development of the QTV server was divided into three phases. First phase involved identifying the requirement of the streaming server and by the end of this phase, being able to handle RTSP requests with appropriate RTSP responses. - Second phase was focused on the development of RTP/RTCP traffic towards the client. - Third phase is dedicated to enhancement of the server with some special features, like Message Drop, SMART and DUMB mode, OOO sequence etc.

Project Name	<i>eNterprise VoIP Phone</i>
Company	Aelix Communication Systems, Bangalore
Duration/ Position	March 2005 to March 2006/Member of Technical Staff
Technology	C, Linux and SIP/SDP protocols

Summary	eNterprise Phone was a SIP (Session Initiation Protocol) based phone. It also supports the extended functionality of SIP protocol i.e. Subscribe and Notify methods defined in RFC3265. By these methods users can subscribe for multiple events and whenever events occurred in SIP Server then registered users will receive notification for specific events. eNterprise phones support the subscription for Date and Time, LED status and On and Off hook status of phones which notify asynchronously to Millennium VoIP line cards and also receive Date and Time updates during phone setup.
Responsibilities	- Involved in design, coding and unit testing of Subscribe and Notify methods in eNterprise phones. - Added similar support in Millennium VoIP line card (Sip Gateway). - Involved in Bug fixes of SIP and RTP/RTCP protocols.

PERSONAL VITAE

Father's Name : Vijay Singh Chandel
 Date of birth : 10th January 1980
 Marital Status : Married
 Nationality : Indian
 Passport Number : P5072746